3.2.2

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Question

Draw a triangle $\triangle ABC$ in which $AB = 5\,\text{cm}$, $BC = 6\,\text{cm}$ and $\angle ABC = 60^\circ$.

Solution

→ Let

$$a = \|\mathbf{C} - \mathbf{B}\| = 6\,\text{cm} \tag{1}$$
$$b = \|\mathbf{A} - \mathbf{C}\| \quad \text{(to be found)} \tag{2}$$
$$c = \|\mathbf{B} - \mathbf{A}\| = 5\,\text{cm} \tag{3}$$

→ By cosine law in $\triangle ABC$ at $\mathbf{B}$,

$$b^2 = a^2 + c^2 - 2ac\cos B \tag{4}$$
$$= 6^2 + 5^2 - 2 \cdot 6 \cdot 5 \cdot \cos 60^\circ \tag{5}$$
$$= 36 + 25 - 60 \cdot \frac{1}{2} \tag{6}$$
$$= 31 \;\Rightarrow\; b = \sqrt{31} \approx 5.57\,\text{cm} \tag{7}$$

Solution

$\rightarrow$ Choose coordinates (placing B at the origin and BC on the x-axis):

$$\mathbf{B} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \tag{8}$$

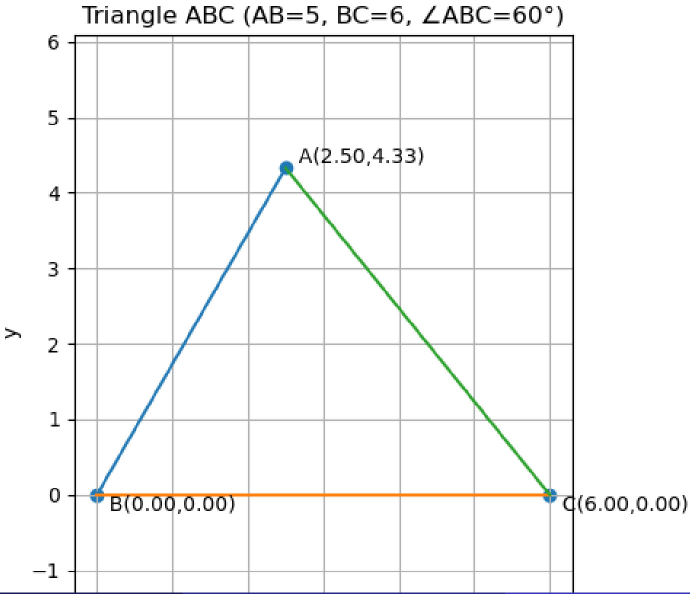
$$\mathbf{C} = \begin{pmatrix} 6 \\ 0 \end{pmatrix} \tag{9}$$

$$\mathbf{A} = c \begin{pmatrix} \cos B \\ \sin B \end{pmatrix} = \begin{pmatrix} \frac{5}{2} \\ \frac{5\sqrt{3}}{2} \end{pmatrix} \approx \begin{pmatrix} 2.5 \\ 4.33 \end{pmatrix} \tag{10}$$

$\rightarrow$ Therefore, the vertices of $\triangle ABC$ are:

$$\mathbf{A} = \begin{pmatrix} 2.5 \\ 4.33 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 6 \\ 0 \end{pmatrix} \tag{11}$$

Solution



Triangle ABC (AB=5, BC=6, ∠ABC=60°)
A(2.50,4.33)
B(0.00,0.00)
C(6.00,0.00)
y
6
5
4
3
2
1
0
−1

C Code

```c
#include <math.h>
int triangle_abc(double AB, double BC, double angleB_deg,
                double* Ax, double* Ay,
                double* Bx, double* By,
                double* Cx, double* Cy,
                double* b_out)
{
    const double pi = 3.14159265358979323846;
    double B = angleB_deg * (pi/180.0);

    // Place B at origin and C on x-axis
    *Bx = 0.0; *By = 0.0;
    *Cx = BC; *Cy = 0.0;

    // From B, place A using polar (AB, angle B)
    *Ax = AB * cos(B);
    *Ay = AB * sin(B);
```

C Code

```c
    // b = AC
    double dx = *Ax - *Cx, dy = *Ay - *Cy;
    *b_out = sqrt(dx*dx + dy*dy);

    return 0;
}
int triangle_example(double* Ax, double* Ay,
                     double* Bx, double* By,
                     double* Cx, double* Cy,
                     double* b_out)
{
    return triangle_abc(5.0, 6.0, 60.0, Ax, Ay, Bx, By, Cx, Cy,
        b_out);
}
```

Python Code

```
import math
import matplotlib.pyplot as plt

AB = 5.0
BC = 6.0
Bdeg = 60.0
Brad = math.radians(Bdeg)

# Coordinates: place B at origin and C on +x as in your LaTeX
Bx, By = 0.0, 0.0
Cx, Cy = BC, 0.0
Ax, Ay = AB*math.cos(Brad), AB*math.sin(Brad) # (2.5, 4.330...)

# Optional: AC to confirm = sqrt(31)
AC = math.hypot(Ax-Cx, Ay-Cy)

# Plot
plt.figure(figsize=(5,5))
```

Python Code

```
    # triangle edges
plt.plot([Ax, Bx], [Ay, By]) # AB
plt.plot([Bx, Cx], [By, Cy]) # BC
plt.plot([Cx, Ax], [Cy, Ay]) # CA
# points
plt.scatter([Ax, Bx, Cx], [Ay, By, Cy])
plt.text(Ax, Ay, f A({Ax:.2f},{Ay:.2f}), ha=left, va=bottom)
plt.text(Bx, By, f B({Bx:.2f},{By:.2f}), ha=left, va=top)
plt.text(Cx, Cy, f C({Cx:.2f},{Cy:.2f}), ha=left, va=top)
```

Python Code

```
    plt.title(Triangle ABC (AB=5, BC=6, ABC=60))
plt.axis(equal)
plt.grid(True)
plt.xlabel(x)
plt.ylabel(y)
plt.tight_layout()
# Save and/or show
plt.savefig(triangle_abc.png, dpi=200)
plt.show()
print(fAC  {AC:.4f} (should be sqrt(31)  5.5678))
```

callc.py

```
import ctypes
from ctypes import c_double, c_int, byref

lib = ctypes.CDLL(./libtriangle.so) # adjust path if needed

# void-return (int), args: 11 doubles (3 inputs + 7 outputs)
lib.triangle_abc.argtypes = [c_double, c_double, c_double,
                             ctypes.POINTER(c_double), ctypes.POINTER(c_double),
                             ctypes.POINTER(c_double), ctypes.POINTER(c_double),
                             ctypes.POINTER(c_double), ctypes.POINTER(c_double),
                             ctypes.POINTER(c_double)]
lib.triangle_abc.restype = c_int
```

callc.py

```
lib.triangle_example.argtypes = [ctypes.POINTER(c_double), ctypes
    .POINTER(c_double),
                                  ctypes.POINTER(c_double), ctypes.
                                      POINTER(c_double),
                                  ctypes.POINTER(c_double), ctypes.
                                      POINTER(c_double),
                                  ctypes.POINTER(c_double)]
lib.triangle_example.restype = c_int

def run_example():
    Ax= c_double(); Ay= c_double()
    Bx= c_double(); By= c_double()
    Cx= c_double(); Cy= c_double()
    b = c_double()

    lib.triangle_example(byref(Ax), byref(Ay), byref(Bx), byref(
        By),
                         byref(Cx), byref(Cy), byref(b))
```

callc.py

```
    print(Example (AB=5, BC=6, B=60))
   print(fA = ({Ax.value:.2f}, {Ay.value:.2f}))
   print(fB = ({Bx.value:.2f}, {By.value:.2f}))
   print(fC = ({Cx.value:.2f}, {Cy.value:.2f}))
   print(fAC = sqrt(31)  {b.value:.2f})

def run_general(AB=5.0, BC=6.0, angleB_deg=60.0):
   Ax= c_double(); Ay= c_double()
   Bx= c_double(); By= c_double()
   Cx= c_double(); Cy= c_double()
   b = c_double()

   lib.triangle_abc(AB, BC, angleB_deg,
```